

Unit-3

Random Processes and

Autocorrelation :-

classification:

1. Stationary process

Note:

when $x(t)$ is a stationary process,

$$R\{t_1, t_2\} = E\{x(t_1)x(t_2)\} \text{ is a function of } (t_1 - t_2)$$

$$\therefore E\{x^2(t)\} \text{ is a constant}$$

Also i) $E\{x(t)\}$ is a constant.

$\therefore$ i) $\text{Var}(x(t))$ is a constant.

wss $\rightarrow$ wide sense stationary process

wss process:

1. $E(x(t))$ is a constant.

2. $R(t_1, t_2) = E(x(t_1)x(t_2))$ is fn of $(t_1 - t_2)$

SSS process:

* $E(x(t))$, $\text{Var}(x(t))$ is a constant

* $R(t_1, t_2) = E(x(t_1)x(t_2))$ is function of $(t_1 - t_2)$

① The process $\{x(t)\}$ whose probability distribution under certain condition is given by.

$$P\{X(t) = n\} = \begin{cases} \frac{(at)^{n-1}}{(1+at)^{n+1}}, & n = 1, 2, 3, \dots \\ \frac{at}{1+at}, & n = 0 \end{cases}$$

show that it is not stationary.

Sol:

check in order:

i) $E(X(t))$ is constant

ii) $\text{Var}(X(t))$ is constant.

iii). $R(t_1, t_2) = E[X(t_1) X(t_2)]$ is constant

Probability mass fn:

$X(t)$

$$P(X(t) = n) = \begin{matrix} 0 & 1 & 2 & 3 & \dots \end{matrix} \quad \begin{matrix} \frac{at}{1+at} & \frac{1}{(1+at)^2} & \frac{at}{(1+at)^3} & \frac{at^2}{(1+at)^4} & \dots \end{matrix}$$

$$E(X(t)) = \sum X(t) P(X(t) = n)$$

$$= 0 + 1 \times \frac{1}{(1+at)^2} + 2 \times \frac{at}{(1+at)^3} + \frac{3 \times (at)^2}{(1+at)^4} + \dots$$

$$= \frac{1}{(1+at)^2} \left[1 + \frac{2at}{(1+at)} + \frac{3(at)^2}{(1+at)^2} + \dots \right]$$

$$= \frac{1}{(1+at)^2} \left[\left(1 - \frac{at}{1+at} \right)^{-2} \right] \quad \left(\frac{1}{1+at} \right)^2$$

$$= \frac{1}{(1+at)^2} \times (1+at)^2 = 1$$

$E(X(t)) = 1$ is a constant

$$E(x^2(t)) = \sum x^2(t) P(x(t)=n)$$

$$= \sum x(1+x) P(x(t)=n) - \sum x P(x(t)=n)$$

$$= 1 \times 2 \cdot \frac{1}{(1+at)^2} + 2 \times 3 \cdot \frac{1}{(1+at)^3} + \dots - (1)$$

$$= \frac{2}{(1+at)^2} \left[1 + \frac{2at}{(1+at)} + \frac{6}{(1+at)^2} + \dots \right] - 1$$

$$= \frac{2}{(1+at)^2} \left(1 - \frac{at}{(1+at)} \right)^{-3} = \frac{(1+at/at)^3}{(1+at)^2}$$

$$= \frac{2}{(1+at)^2} (1+at)^3 - 1$$

$$E(x^2(t)) = 2(1+at) - 1$$

$$E(x^2(t)) = 1 + 2at \text{ is not constant.}$$

$\therefore P(x(t)=n)$ is not stationary Process.

② If $x(t)$ is a wide sense stationary process (WSS) with auto correlation R

③ show that the random process $x(t) = A \cos(\omega_0 t + \theta)$ is wide sense stationary.

If A & ω_0 are constants and θ is uniform distributed random variable in $(0, 2\pi)$

Sol:

$$f(\theta) = \frac{1}{2\pi}, \quad 0 < \theta < 2\pi$$

$$E(x(t)) = \int x(t) f(\theta) d\theta$$

$$= \int_0^{2\pi} A \cos(\omega_0 t + \theta) \frac{1}{2\pi} d\theta$$

$\theta=0$

$$\begin{aligned}
 &= \frac{A}{2\pi} \int_0^{2\pi} \cos(\omega_0 t + \theta) d\theta \\
 &= \frac{A}{2\pi} \left[\sin(\omega_0 t + \theta) \right]_0^{2\pi} \\
 &= \frac{A}{2\pi} (\sin(\omega_0 t + 2\pi) - \sin(\omega_0 t)) \\
 &= \frac{A}{2\pi} (\sin \omega_0 t - \sin \omega_0 t)
 \end{aligned}$$

$E(x(t)) = 0$ is a constant.

$$R(t_1, t_2) = E(x(t_1)x(t_2))$$

$$\begin{aligned}
 &= \int_0^{2\pi} A \cos(\omega_0 t_1 + \theta) \cdot A \cos(\omega_0 t_2 + \theta) f(\theta) d\theta \\
 &= \frac{A^2}{2\pi \times 2} \int_0^{2\pi} [\cos(\omega_0(t_1 + t_2) + 2\theta) + \cos(\omega_0(t_1 - t_2))] d\theta
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{A^2}{4\pi} \int_0^{2\pi} \left[\frac{\sin(\omega_0(t_1 + t_2) + 2\theta)}{2} + \cos(\omega_0(t_1 - t_2)) \theta \right] d\theta
 \end{aligned}$$

$$\begin{aligned}
 \sin(2\pi + \theta) &= \sin(\theta) \\
 &\Rightarrow \sin(\theta)
 \end{aligned}$$

$$\begin{aligned}
 &= \frac{A^2}{4\pi} \left[\frac{\sin(\omega_0(t_1 + t_2) + 2\theta)}{2} + 2\pi \cos \omega_0(t_1 - t_2) \right]_0^{2\pi} \\
 &= \frac{A^2}{4\pi} (2\pi \cos \omega_0(t_1 - t_2))
 \end{aligned}$$

$$R(t_1, t_2) = \frac{A^2}{2} (\cos \omega_0(t_1 - t_2))$$

$\therefore R(t_1, t_2)$ is a fn of $(t_1 - t_2)$

$x(t)$ is a WSS process

③ Given a RV Y with characteristic fn

$$\phi(\omega) = E\{e^{i\omega Y}\} = E\{\cos \omega Y + i \sin \omega Y\} \text{ and}$$

a random process defined by $x(t) = \cos(\lambda t + Y)$
show that $\{x(t)\}$ is a stationary in wide sense if $\phi(1) = \phi(2) = 0$.

sol:

$$E(x(t)) = E[\cos(\lambda t + Y)]$$

$$= E[\cos \lambda t \cos Y - \sin \lambda t \sin Y]$$

$$E(x(t)) = \cos \lambda t E[\cos Y] - \sin \lambda t E[\sin Y] \quad \hookrightarrow \textcircled{a}$$

$$\phi(\omega) = E[\cos \omega Y + i \sin \omega Y]$$

$$\phi(\omega) = E[\cos \omega Y] + i E[\sin \omega Y]$$

$$\Rightarrow \phi(1) = E[\cos Y] + i E[\sin Y]$$

$$\phi(1) = 0 + i0 = E[\cos Y] + i E[\sin Y]$$

equating on b/s.

$$E[\cos Y] = 0$$

$$E[\sin Y] = 0$$

① $\Rightarrow$

$$E[x(t)] = \cos \lambda t (0) - \sin \lambda t (0)$$

$E(x(t)) = 0$ is a constant.

$$R(t_1, t_2) = E[x(t_1)x(t_2)]$$

$$= E[\cos(\lambda t_1 + Y) \cos(\lambda t_2 + Y)]$$

$$= E\left[\frac{\cos(\lambda(t_1 + t_2) + 2Y) + \cos \lambda(t_1 - t_2)}{2}\right]$$

$$\begin{aligned}
&= E \left[(\cos \lambda t_1, \cos \gamma - \sin \lambda t_1, \sin \gamma) (\cos \lambda t_2 \cos \gamma - \sin \lambda t_2 \sin \gamma) \right] \\
&= \cos \lambda t_1 \cos \lambda t_2 E[\cos \gamma] - \cos \lambda t_1 \sin \lambda t_2 E[\cos \gamma \sin \gamma] \\
&\quad - \sin \lambda t_1 \cos \lambda t_2 E[\cos \gamma \sin \gamma] + \sin \lambda t_1 \sin \lambda t_2 E[\sin^2 \gamma] \\
&= \frac{1}{2} \{ \cos \lambda t_1 \cos \lambda t_2 + \cos \lambda t_1 \cos \lambda t_2 E[\cos 2\gamma] \} - \\
&\quad \frac{1}{2} \{ E(\sin 2\gamma) (\cos \lambda t_1 \sin \lambda t_2 + \sin \lambda t_1 \cos \lambda t_2) \} \\
&\quad + \frac{1}{2} \{ \sin \lambda t_1 \sin \lambda t_2 - \sin \lambda t_1 \sin \lambda t_2 E[\cos 2\gamma] \}.
\end{aligned}$$

$$E[\cos 2\gamma] = E[\sin 2\gamma] = 0.$$

$$\begin{aligned}
&= \frac{1}{2} \{ \cos \lambda t_1 \cos \lambda t_2 + \sin \lambda t_1 \sin \lambda t_2 \} \\
&= \frac{1}{2} \{ \cos (\lambda t_1 - \lambda t_2) \} \\
&= \frac{1}{2} \cos \lambda (t_1 - t_2)
\end{aligned}$$

$$\therefore R(t_1, t_2) \text{ is a fn of } (t_1 - t_2)$$

$$\therefore X(t) \text{ is WSS.}$$

④ In the fair coin experiment we define the process $\{X(t)\}$ as follows:

$$X(t) = \sin \pi t, \text{ if head shows and}$$

$$= 2t, \text{ if tail shows.}$$

a). Find $E\{X(t)\}$ b). Find $F(x, t)$ for $t = 0.25$

Sol:-

The probability distribution function $x(t)$

$x(t)$	$\sin \pi t$	$2t$
$P(x(t)=n)$	$1/2$	$1/2$

$$E[x(t)] = \sum x(t) P(x(t)=n)$$

$$= \sin \pi t \times \frac{1}{2} + 2t \times \frac{1}{2}$$

$$E[x(t)] = \frac{\sin \pi t}{2} + t //$$

at $t = 0.25 = 1/4 \Rightarrow \sin \pi/4 = 1/\sqrt{2}$, $2t = 2 \times \frac{1}{4} = 1/2$

$x(t)$	$\frac{1}{\sqrt{2}}$	$\frac{1}{2}$
$P(x(t)=n)$	$\frac{1}{2}$	$1/2$

$$F(x, t) = \begin{cases} 0 & -\infty < x < 1/\sqrt{2} \\ \frac{1}{2} & 1/\sqrt{2} < x < 1/2 \\ 1 & 1/2 < x < +\infty \end{cases}$$

Q. If $x(t)$ is a WSS process with autocorrelation $R(\tau) = Ae^{-\alpha|\tau|}$, determine the second order moment of RV $x(8) - x(5)$

Sol: $R(z) = R(t_1, t_2) = E[x(t_1)x(t_2)] = A e^{-\alpha|t_1 - t_2|}$

$$E[(x(8) - x(5))^2] = E[x^2(8)] + E[x^2(5)] - 2E[x(8)x(5)]$$

if $x(t_1) = x(t_2)$ [$\because e^0 = 1$]

$$E[x^2(t_1)] = A \cdot e^{-\alpha(0)} = A$$

$$E[x^2(8)] = E[x^2(5)] = A$$

$$E[x(8)x(5)] = A e^{-\alpha|8-5|} = A e^{-3\alpha}$$

$$E[(x(8) - x(5))^2] = A + A - 2A e^{-3\alpha} = 2A - 2A e^{-3\alpha}$$

$$= 2A(1 - e^{-3\alpha})$$

6. show that the process $x(t) = A \cos \lambda t + B \sin \lambda t$ where A and B are RVs is a WSS process if

i). $E(A) = E(B) = 0$.

ii). $E(A^2) = E(B^2)$

iii). $E(AB) = 0$.

Sol:

$$E[x(t)] = E(A \cos \lambda t + B \sin \lambda t)$$

$$E[x(t)] = \cos \lambda t E(A) + \sin \lambda t E(B)$$

$$\therefore E(A) = E(B) = 0$$

$$E[x(t)] = 0.$$

$$R(t_1, t_2) = E[x(t_1)x(t_2)]$$

$$= E[(A \cos \lambda t_1 + B \sin \lambda t_1)(A \cos \lambda t_2 + B \sin \lambda t_2)]$$

$$= \cos \lambda t_1 \cos \lambda t_2 E(A^2) + \cos \lambda t_1 \sin \lambda t_2 E[AB]$$

$$+ \sin \lambda t_1 \cos \lambda t_2 E(AB) + \sin \lambda t_1 \sin \lambda t_2 E(B^2)$$

$$\text{Let } E(A^2) = E(B^2) = K.$$

$$= K [\cos \lambda t_1 \cos \lambda t_2 + \sin \lambda t_1 \sin \lambda t_2]$$

$$= K [\cos(\lambda t_1 - \lambda t_2)]$$

$$R(t_1, t_2) = K [\cos \lambda (t_1 - t_2)]$$

$$\therefore R(t_1, t_2) \text{ is a fn of } (t_1 - t_2)$$

$\therefore x(t)$ is a WSS process.

- ⑦. Given a RV ω with density $f(\omega)$ and another RV ϕ uniformly distributed in $(-\pi, \pi)$ and independent of ω and $x(t) = a \cos(\omega t + \phi)$ prove that $x(t)$ is a WSS process.

- ⑧. Verify whether the sinewave process $\{x(t)\}$ where $x(t) = \gamma \cos \omega t$, where γ is uniformly distributed in $(0, 1)$ is a SSS process.

Solt

$$F_X(x(t)) = P(X(t) \leq x)$$

$$\Rightarrow P(Y \cos \omega t \leq x)$$

$$= \begin{cases} P\left(Y \leq \frac{x}{\cos \omega t}\right) & , \cos \omega t > 0 \\ P\left(Y \geq \frac{x}{\cos \omega t}\right) & , \cos \omega t < 0 \end{cases}$$

$$F_X(x(t)) = \begin{cases} F\left(Y \leq \frac{x}{\cos \omega t}\right) & , \cos \omega t > 0 \\ 1 - F\left(Y < \frac{x}{\cos \omega t}\right) & , \cos \omega t < 0 \end{cases}$$

$$f_X(x(t)) = f\left(\frac{x}{\cos \omega t}\right) \cdot \frac{1}{\cos \omega t}$$

It is fn of t.

$\therefore X(t)$ is SSS process.

(7) Sol: Given, $f(\phi) = \frac{1}{2\pi}$, $(-\pi < \phi < \pi)$

$$x(t) = a \cos(\omega t + \phi)$$

$$E(x(t)) = \int x(t) f(\phi) d\phi.$$

$$= \int_{-\pi}^{\pi} a \cos(\omega t + \phi) \cdot \frac{1}{2\pi} d\phi.$$

$$= \frac{a}{2\pi} \left[\sin(\omega t + \phi) \right]_{-\pi}^{\pi}$$

$$= \frac{a}{2\pi} \left[\sin(\omega t + \pi) - \sin(\omega t - \pi) \right]$$

$$= \frac{a}{2\pi} \left[-\sin \omega t - (-\sin \omega t) \right]$$

$$= 0.$$

$\therefore E(x(t)) = 0$ is a constant.

$$R(t_1, t_2) = E[x(t_1)x(t_2)]$$

$$= \int a \cos(\omega t_1 + \phi) \cdot a \cos(\omega t_2 + \phi) \cdot \frac{1}{2\pi} d\phi.$$

$$= \frac{a^2}{2\pi \times 2} \int_{-\pi}^{\pi} \cos(\omega(t_1 + t_2) + 2\phi) + \cos(\omega(t_1 - t_2)) d\phi.$$

$$= \frac{a^2}{4\pi} \left[\frac{\sin(\omega(t_1 + t_2) + 2\phi)}{2} + \cos(\omega(t_1 - t_2)) \right]_{-\pi}^{\pi}$$

$$\sin(\theta + 2\pi) = \sin \theta$$

$$\sin(\theta - 2\pi) = \sin \theta$$

$$= \frac{a^2}{4\pi} \left[\frac{\sin(\omega(t_1 + t_2) + 2\pi)}{2} + \pi \cos \omega(t_1 - t_2) \right.$$

$$\left. - \frac{\sin(\omega(t_1 + t_2) - 2\pi)}{2} + \pi \cos \omega(t_1 - t_2) \right]$$

$$= \frac{a^2}{4\pi} (2\pi \cos \omega(t_1 - t_2))$$

$$R(t_1, t_2) = \frac{\sigma^2}{2} \cos(t_1 - t_2)$$

$\therefore R(t_1, t_2)$ is a fn of $(t_1 - t_2)$

$X(t)$ is WSS process.

$$E\{X(t)X(t+\tau)\} = R(\tau)$$

$$E\{X(t)X(t+\tau)\} = \frac{\sigma^2}{2} \cos(\tau)$$

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Poisson Process:

If $x(t)$ represents no. of occurrence of certain events in $(0, t)$, then the discrete random process $x(t)$ is called poisson process.

probability law for poisson process.

$$P_n(t) = P(x(t) = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}, n = 0, 1, 2, \dots$$

$$\text{Mean} = \lambda t$$

$$\text{Variance} = \lambda t.$$

$\lambda \rightarrow$ mean arrival rate
 $t \rightarrow$ time

Properties of Poisson Process:

1. The poisson process is a Markov process.

2. Sum of two independent poisson process is a poisson process $\checkmark$ (with proof)

3. difference of two independent poisson process is not a poisson process.

Note:

$\Rightarrow$ consecutive arrivals follows exponential distribution

$$f(t) = \lambda e^{-\lambda t}, t \geq 0.$$

$$\text{If Given } \lambda, t, P \Rightarrow \text{then } P_n(t) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}, n = 0, 1, 2, \dots$$

- Q. suppose that customers arrive at a bank according to a poisson process with a mean rate of 3 per minute. Find the probability that during a time interval two minutes
- exactly 4 customers arrived.
 - More than 4 customers arrived.

Sol: Mean arrival rate, $\lambda = 3$
 $t = 2$.

$$\lambda t = 3 \times 2 = 6.$$

$$P(X(t) = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}$$

$$\begin{aligned} \text{i). } P(X(2) = 4) &= \frac{e^{-6} (6)^4}{4!} = \frac{0.478 \times 10^{-3} \times 1296}{24} \\ &= \frac{3.2124}{24} \end{aligned}$$

$$= 0.1338.$$

$$\text{ii)} \quad P(X(2) > 4) = 1 - P(X(2) \leq 4)$$

$$= 1 - \{ P(X(2) = 0) + P(X(2) = 1) + P(X(2) = 2) + P(X(2) = 3) + P(X(2) = 4) \}$$

$$= 1 - \left\{ e^{-6} + e^{-6} \times 6 + \frac{e^{-6} (6)^2}{2} + \frac{e^{-6} (6)^3}{6} + \frac{e^{-6} (6)^4}{24} \right\}$$

$$= 1 - e^{-6} \{ 1 + 6 + 18 + 36 + 54 \}.$$

$$= 1 - e^{-6} \{ 115 \}.$$

$$= 1 - 0.2478 \times 10^{-3} \times 115.$$

$$= 1 - 0.2850$$

$$= 0.71494.$$

$$P(X(2) > 4) \approx 0.715$$

(10). A machine goes out of order whenever a component fails, the failure of this part follow the poisson process with a mean rate of 1 per week. Find the probability that 2 weeks have elapsed since last failure. If there are five spare parts of this component in an inventory, and that the next supply is not due in 10 weeks. Find the probability that the machine will not be out of order in the next 10 weeks.

Sol:

Given,

$$\lambda = 1.$$

i). time = 2 weeks (No failure)

$$\lambda t = 1 \times 2 = 2.$$

$$P(X(2) = 0) = \frac{e^{-2} (2)^0}{0!}$$

$$= 0.135$$

ii). $n = 5$ (space parts)

$$t = 10$$

$$\lambda t = 10$$

$$P(X(10) \leq 5) = P(X(10) = 0) + P(X(10) = 1) + \\ P(X(10) = 2) + P(X(10) = 3) + P(X(10) = 4) + \\ P(X(10) = 5)$$

$$= e^{-10} + e^{-10} \cdot 10 + \frac{e^{-10} \cdot (10)^2}{2} + \frac{e^{-10} \cdot (10)^3}{6} \\ + \frac{e^{-10} \cdot (10)^4}{24} + \frac{e^{-10} \cdot (10)^5}{120}$$

$$= e^{-10} \left[1 + 10 + 50 + \frac{1000}{6} + 416.66 + 833.33 \right]$$

$$= e^{-10} [1477.65]$$

$$= 4.539 \times 10^{-5} \times 1477.65$$

$$= 0.06708$$

⑥. If customers arrive at a counter in a poisson process with a mean rate of two per minutes. Find the probability that the interval btw 2 consecutive arrivals is

i). More than 1 minute.

ii). btw 1 & 2 minutes.

iii). 4 Minute (or) less than equal.

Sol:-

consecutive arrival follows exponential

$$f(t) = \lambda e^{-\lambda t}, \quad t \geq 0.$$

$$\therefore \lambda = 2$$

$$f(t) = 2e^{-2t}, \quad t \geq 0.$$

$$\Rightarrow P(T > 1) = \int_1^{\infty} f(t) dt.$$

$$= \int_1^{\infty} 2e^{-2t} dt.$$

$$= 2 \left[\frac{e^{-2t}}{-2} \right]_1^{\infty}$$

$$= -[0 - e^{-2}]$$

$$= 0.1353.$$

$$\text{ii) } P(1 < T < 2) = \int_1^2 f(t) dt.$$

$$= \int_1^2 2e^{-2t} dt.$$

$$= 2 \left[\frac{e^{-2t}}{-2} \right]_1^2$$

$$= -[e^{-4} - e^{-2}]$$

$$= 0.135 - 0.018.$$

$$= 0.1170.$$

$$\text{iii) } P(T \leq 4) = \int_0^4 f(t) dt.$$

$$= 2 \left[\frac{e^{-2t}}{-2} \right]_0^4$$

$$= -[e^{-8} - 1]$$

$$= 1 - e^{-8}$$

$$= 1 - 3.354 \times 10^{-4}$$

$$= 0.999.$$

- ⑫. A radio active source emits particles at a rate of 5 per minutes in with poisson process each particle emitted as probability 0.6 of being recorded. Find the probability that 10 particles are recorded in 4 mins period.

Sol:-

$$\lambda = 5$$

$$P = 0.6$$

$$n = 10.$$

$$t = 4, \quad P \lambda t = 5 \times 4 = 20 \times 0.6$$

$$P(X(4) = 10) = \frac{e^{-20} (20)^{10}}{10!} = \frac{e^{-12} (12)^{10}}{10!}, \quad t \geq 0.$$

$$\left[e^{-12} + e^{-12} (12) + \frac{e^{-12} (12)^2}{2} + \frac{e^{-12} (12)^3}{6} + \frac{e^{-12} (12)^4}{24} + \frac{e^{-12} (12)^5}{120} + \frac{e^{-12} (12)^6}{720} \right]$$

$$= \frac{6.1442 \times 10^{-6} \times 6.1917 \times 10^{10}}{10!}$$

$$= \frac{380433.4342}{3628800}$$

$$= 0.1048$$

13. A radioactive source emits particles at a rate of 6 per minute in poisson process each particle emitted as a probability of $\frac{1}{3}$ of being recorded. Find the probability that atleast 5 particles are recorded in a 5 min period.

Sol:-

Given, $\lambda = 6$.

$P = \frac{1}{3}$.

$t = 5$

$n \leq 5$

$$P\lambda t = \frac{1}{3} \times 6 \times 5 = 10.$$

$$P(X(5) \leq 5) = \frac{e^{-10} (10)^n}{n!}$$

$$= P(X(5)=0) + P(X(5)=1) + P(X(5)=2) + P(X(5)=3) + P(X(5)=4) + P(X(5)=5)$$

$$= e^{-10} + e^{-10}(10) + \frac{e^{-10}(10)^2}{2} + \frac{e^{-10}(10)^3}{6} + \frac{e^{-10}(10)^4}{24}$$

$$+ \frac{e^{-10}(10)^5}{120}$$

$$= e^{-10} [1 + 10 + 50 + 166.66 + 416.66 + 833.33]$$

$$= 4.539 \times 10^5 \times 1477.65$$

$$= 0.66708$$

Auto Correlation fn and it's properties.

Def:

If the process $x(t)$ is stationary either in the SSS (or) in the WSS, then

$E[x(t)x(t+\tau)]$ is fn of τ is denoted by

$$R_{xx}(\tau) \text{ (or) } R_x(\tau) \text{ (or) } R(\tau).$$

This function $R(\tau)$ is called the auto correlation function of the process $x(t)$.

Note:

$$R_{xx}(\tau) = E[x(t)x(t+\tau)]$$

$$= E[x(t+\tau)x(t)]$$

$$= E[x(t)x(t-\tau)]$$

$$= E[x(t-\tau)x(t)]$$

Properties:

1) $R(\tau)$ is an even function of τ

proof: $R_{xx}(\tau) = E[x(t)x(t+\tau)] = R(\tau)$

$$R(-\tau) = E[x(t)x(t-\tau)]$$

$$R(-\tau) = R(\tau)$$

$\therefore R(\tau)$ is even fn

2) $R(\tau)$ is maximum at $\tau=0$ that is

$$|R(\tau)| \leq R(0)$$

proof: Squats Inequality $\rightarrow [E(xy)]^2 \leq E(x^2)E(y^2)$

$$(E[x(t)x(t+\tau)])^2 \leq E[x^2(t)] E[x^2(t+\tau)]$$

$$\therefore R(\tau) = E[x(t)x(t+\tau)]$$

$$R(0) = E[x(t)x(t+0)]$$

$$R(0) = E[x^2(t)]$$

$$[R(\tau)]^2 \leq R(0) R(0)$$

$$(R(\tau))^2 \leq (R(0))^2$$

take sq rt,

$$R(\tau) \leq R(0)$$

$$|R(\tau)| \leq R(0).$$

3b. If the autocorrelation function $R(t)$ of a Real stationary process $x(t)$ is continuous at $T=0$. It is continuous at every other point.

Proof: $R(t)$ is continuous at $T=0$.

$$\text{consider, } E[x(t) - x(t-\tau)]^2 = E[x^2(t)] + E[x^2(t-\tau)]$$

$$- 2E[x(t)x(t-\tau)]$$

$$= R(0) + R(0) - 2R(\tau)$$

$$E[x(t) - x(t-\tau)]^2 = 2(R(0) - R(\tau))$$

$$\lim_{T \rightarrow 0} \text{LHS} = 0 \quad E[x(t) - x(t)]^2 = 0$$

$$\lim_{T \rightarrow 0} \text{RHS} = 0 \quad 2(R(0) - R(0)) = 0$$

$$\text{LHS} = \text{RHS}$$

$\therefore R(\tau)$ is continuous at $T=0$.

consider, $R(\tau+h) - R(\tau) = E[x(t) x(t+\tau+h)] -$

$$E[x(t) x(t+\tau)]$$

$$R(\tau+h) - R(\tau) = E[x(t)] \{ E[x(t+\tau+h)] - E[x(t+\tau)] \}$$

$$\lim_{h \rightarrow 0} \text{LHS} = 0 \quad R(\tau+0) - R(\tau) = 0$$

$$\lim_{h \rightarrow 0} \text{RHS} = 0 \quad E[x(t+\tau+0)] - E[x(t+\tau)] = 0$$

$$\text{LHS} = \text{RHS}$$

$R(\tau)$ is continuous at other values.

Cross correlation function and its property.

def:

If the process $x(t)$ and $y(t)$ are jointly WSS then $E[x(t) y(t+\tau)]$ is a function of τ denoted by $R_{xy}(\tau)$.

This function $R_{xy}(\tau)$ is called cross correlation function of the process of $x(t)$ and $y(t)$.

properties:

$$1). R_{xy}(\tau) = R_{yx}(-\tau)$$

proof:

$$R_{xy}(\tau) = E[x(t) y(t+\tau)]$$

$$R_{yx}(-\tau) = E[y(t) x(t-\tau)]$$

$$= E[Y(1-T) X(t)]$$

$$R_{xy}(T) = R_{xy}(T)$$

$$\Rightarrow |R_{xy}(T)| \leq \sqrt{R_{xx}(0) R_{yy}(0)}$$

Proof:

$$\text{Inequality, } E[XY]^2 \leq E(X^2) E(Y^2)$$

$$E[X(t) Y(t+T)]^2 \leq E[X^2(t)] E[Y^2(t+T)]$$

$$\therefore R_{xx}(0) = E[X^2(t)]$$

$$R_{yy}(0) = E[Y^2(t+T)]$$

$$[R_{xy}(T)]^2 \leq R_{xx}(0) \cdot R_{yy}(0)$$

take sqrt on b/s.

$$|R_{xy}(T)| \leq \sqrt{R_{xx}(0) \cdot R_{yy}(0)}$$

$$3\}. |R_{xy}(T)| \leq \frac{1}{2} \{R_{xx}(0) + R_{yy}(0)\}$$

Proof:

Geometric mean $\leq$ Arithmetic mean.

$$\sqrt{ab} \leq \frac{a+b}{2}$$

$$a = X(t), \quad b = Y(t+T)$$

$$\sqrt{X(t) Y(t+T)} \leq \frac{1}{2} [X(t) + Y(t+T)]$$

$$X(t) Y(t+T) \leq \frac{1}{4} [X^2(t) + Y^2(t+T) + 2X(t) Y(t+T)]$$

apply exponential,

$$R_{xy}(T) \leq \frac{1}{4} [R_{xx}(0) + R_{yy}(0) + 2R_{xy}(T)]$$

$$R_{xy}(T) - \frac{1}{2} R_{xy}(T) \leq \frac{1}{4} \{ R_{xx}(0) + R_{yy}(0) \}$$

$$\frac{1}{2} R_{xy}(T) \leq \frac{1}{4} \{ R_{xx}(0) + R_{yy}(0) \}$$

$$|R_{xy}(T)| \leq \frac{1}{2} \{ R_{xx}(0) + R_{yy}(0) \}$$

- (15) A random process $x(t)$ defined as $x(t) = A \cos t + B \sin t$ for all t where A & B are independent binary random variables each of which assume values -1 and 1 with probabilities $\frac{1}{3}$ and $\frac{2}{3}$ respectively. Prove that $\{x(t)\}$ is WSS.

Sol:-

$$x(t) \rightarrow \begin{matrix} 1 \\ 2/3 \end{matrix}$$

$$P(x(t)=-1) \rightarrow \frac{1}{3}$$

to prove $x(t)$ is WSS.

i) $E[x(t)]$ is constant.

ii) $R[t_1, t_2] = E[x(t_1)x(t_2)]$ is fn of (t_1, t_2)

$$\begin{aligned} E[x(t)] &= E[A \cos t + B \sin t] \\ &= \cos t E[A] + \sin t E[B] \end{aligned}$$

$$E[A] = -1 \cdot \frac{1}{3} + 1 \cdot \frac{2}{3} = 0$$

$\therefore A$ & B are independent so

$$E[A] = E[B]$$

$E[x(t)] = 0$ is a constant.

$$R(t_1, t_2) = E[X(t_1)X(t_2)]$$

$$= E[(A \cos t_1 + B \sin t_1)(A \cos t_2 + B \sin t_2)]$$

$$= \cos t_1 \cos t_2 E[A^2] + E[AB] \{ \cos t_1 \overset{\sin}{\cancel{\cos}} t_2 + \sin t_1 \cos t_2 \} + \sin t_1 \sin t_2 E[B^2]$$

$$E[A^2] = 4 \times \frac{1}{3} + 1 \times \frac{2}{3} = \frac{6}{3} = 2 = E[B^2]$$

$$E[AB] = E(A)E(B) = 0$$

$$R(t_1, t_2) = 2 [\cos t_1 \cos t_2 + \sin t_1 \sin t_2]$$

$$R(t_1, t_2) = 2 \cos(t_1 - t_2) \text{ is a fn of } t_1 - t_2.$$

$\therefore X(t)$ is WSS.

- (15) Two random process $X(t)$ and $Y(t)$ are given by $X(t) = A \cos(\omega t + \theta)$, $Y(t) = A \sin(\omega t + \theta)$ where A and ω are constants and θ is a uniform random variable over 0 to 2π . Find the cross-correlation & auto-correlation fn.

Sol:

$$f(\theta) = \frac{1}{2\pi}, \quad 0 < \theta < 2\pi$$

auto correlation of $X(t)$.

$$R_{XX}(\tau) = R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$$

$$= E[(A \cos(\omega t + \theta))(A \cos(\omega t + \theta))]$$

$$= \int_0^{2\pi} A^2 \cos(\omega t + \theta) \cdot \cos(\omega t + \theta) \cdot \frac{1}{2\pi} d\theta$$

$$= \frac{A^2}{4\pi} \int_0^{2\pi} \cos \omega(t_1 + t_2) + 2\theta + \cos \omega(t_1 - t_2) d\theta$$

$$= \frac{A^2}{4\pi} \left[\frac{\sin(\omega(t_1 + t_2) + 2\theta)}{2} + \cos \omega(t_1 - t_2) \theta \right]_0^{2\pi}$$

$$= \frac{A^2}{4\pi} \left[\frac{\sin(\omega(t_1 + t_2) + 4\pi)}{2} + \cos \omega(t_1 - t_2) 2\pi \right.$$

$$\left. - \frac{\sin(\omega(t_1 + t_2))}{2} - 0 \right]$$

$$R_{xx}(t) = R(t_1, t_2) = \frac{A^2}{2} \cos \omega(t_1 - t_2)$$

Autocorrelation of $y(t)$

$$R_{yy}(\tau) = E[y(t_1) y(t_2)]$$

$$= E[A \sin(\omega t_1 + \theta) \cdot A \sin(\omega t_2 + \theta)]$$

$$= \int_0^{2\pi} A^2 \sin(\omega t_1 + \theta) \cdot \sin(\omega t_2 + \theta) \cdot \frac{1}{2\pi} d\theta$$

$$= \frac{A^2}{4\pi} \int_0^{2\pi} \cos \omega(t_1 - t_2) - \cos(\omega(t_1 + t_2) + 2\theta) d\theta$$

$$= \frac{A^2}{4\pi} \left[\frac{\sin \omega(t_1 - t_2) \theta}{2} - \frac{\sin[\omega(t_1 + t_2) + 2\theta]}{2} \right]_0^{2\pi}$$

$$= \frac{A^2}{4\pi} \left[2\pi \cos \omega(t_1 - t_2) - \frac{\sin[\omega(t_1 + t_2) + 4\pi]}{2} \right.$$

$$\sin(\theta + 4\pi) = \sin \theta$$

$$\left. - 0 + \frac{\sin \omega(t_1 + t_2)}{2} \right]$$

$$R_{yy}(\tau) = \frac{A^2}{2} \cos \omega(t_1 - t_2)$$

Cross - correlation

$$R_{xy}(T) = E[x(t_1) y(t_2)]$$

$$= E[A \cos(\omega t_1 + \theta) \cdot A \sin(\omega t_2 + \theta)]$$

$$= \frac{A^2}{2} E[\sin(\omega(t_1 + t_2) + 2\theta) - \sin(\omega(t_1 - t_2))]$$

$$= \frac{A^2}{4\pi} \int_0^{2\pi} \sin \omega(t_1 + t_2) + 2\theta - \sin \omega(t_1 - t_2) d\theta$$

$$= \frac{A^2}{4\pi} \int_0^{2\pi} \left[\frac{-\cos \omega(t_1 + t_2) + 2\theta}{2} - \sin \omega(t_1 - t_2) \theta \right] d\theta$$

$$= \frac{A^2}{4\pi} \left[\frac{-\cos [\omega(t_1 + t_2) + 4\pi]}{2} - 2\pi \sin \omega(t_1 - t_2) \right]$$

$$= \frac{A^2}{4\pi} \left[\frac{-\cos \omega(t_1 + t_2) + 0}{2} + \frac{\cos \omega(t_1 + t_2)}{2} + 0 \right]$$

$$= \frac{A^2}{4\pi} (-2\pi \sin \omega(t_1 - t_2))$$

$$R_{xy}(T) = \frac{A^2}{2} \sin \omega(t_2 - t_1)$$

Random Telegraph Signal Process:

(it follows poisson process).

Def:

Let $\{N(t); t \geq 0\}$ be a poisson process and α be another Random variables which is independent of $N(t)$ such that.

$$\alpha = -1$$

$$P(\alpha) = \frac{1}{2} \quad \frac{1}{2}, \text{ then } x(t) = \alpha(-1)^{N(t)}$$

is called random telegraph signal process.

Semi Random Telegraph Signal Process:

Def: If $N(t)$ represents a number of occurrence of a specified even in $(0, t)$ and $x(t) = (-1)^{N(t)}$, then $x(t)$ is called semi-Random Telegraph signal process.

(16) Show that the random telegraph signal process is WSS.

Sol:

RTP, $x(t) = \alpha (-1)^{N(t)}$

α	-1	1
$P(\alpha)$	$\frac{1}{2}$	$\frac{1}{2}$

To find $x(t)$ is WSS:

(i.e) i) $E[x(t)]$ is a constant.

ii) $R(t_1, t_2)$ is fn of $(t_1 - t_2)$

i) $E[x(t)] = E[\alpha (-1)^{N(t)}]$

$$= E[\alpha] E[(-1)^{N(t)}]$$

$$= E[\alpha] (-1)^{N(t)}$$

$$\therefore E(\alpha) = (-1) \cdot \frac{1}{2} + 1 \cdot \left(\frac{1}{2}\right) = 0$$

$$E[x(t)] = 0 \text{ is a constant.}$$

$$\begin{aligned}
 \text{ii). } R(t_1, t_2) &= E[X(t_1) X(t_2)] \\
 &= E[\alpha^{(-1)^{N(t_1)}} \alpha^{(-1)^{N(t_2)}}] \\
 &= E[\alpha^2 (-1)^{N(t_1) + N(t_2)}]
 \end{aligned}$$

$$\text{ii). } R_{xx}(\tau) = E[X(t) X(t+\tau)]$$

$$= E[\alpha^{(-1)^{N(t)}} \alpha^{(-1)^{N(t+\tau)}}]$$

$$= E[\alpha^2 (-1)^{N(t) + N(t) + N(t) + N(t+\tau)}]$$

$$= E[\alpha^2] (-1)^{2N(t)} (-1)^{N(t+\tau)}$$

power is even so +ve 1.

$$E[\alpha^2] = (-1)^2 \cdot \frac{1}{2} + (1)^2 \cdot \frac{1}{2}$$

$$= 1$$

$$R_{xx}(\tau) = (1) \cdot (1) (-1)^{N(\tau)}$$

$R_{xx}(\tau)$ is 1/h of τ (or) difference

btw t_1 and t_2 .

$X(t)$ is WSS.

(17) Show that semi-random telegraph signal process is WSS evolutionary that is it not SSS and is not a WSS process.

Sol:- by SRTP, $X(t) = (-1)^{N(t)}$

to find: $X(t)$ is SSS.

i). $E[x(t)]$ is constant.

ii). $\text{Var}[x(t)]$ is constant.

$$E[x(t)] = E[(-1)^{N(t)}]$$

$$= (-1)^{N(t)} \text{ is not a constant}$$

$\therefore x(t)$ is not SSS and $x(t)$ is not WSS.

Hence $x(t)$ is nonstationary.

(or)

$x(t)$ is Poisson process.

$$x(t) = (-1)^{N(t)} \begin{cases} \text{even} \\ \text{odd} \end{cases}$$

$$P(x(t) = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!} \quad n = 0, 1, 2, 3, 4, \dots$$

$$P(x(t) = \text{even}) = e^{-\lambda t}$$

$$E[x(t) = \text{even}] = \sum (x(t) = \text{even}) \cdot P(x(t) = n)$$

$$= e^{-\lambda t} + e^{-\lambda t} \frac{(\lambda t)^2}{2!} + e^{-\lambda t} \frac{(\lambda t)^4}{4!} + \dots$$

$$= e^{-\lambda t} \left[\frac{(\lambda t)^0}{0!} + \frac{(\lambda t)^2}{2!} + \frac{(\lambda t)^4}{4!} + \dots \right]$$

$$= e^{-\lambda t} \cosh \lambda t$$

$$E[x(t) = \text{odd}] = e^{-\lambda t} \left[\frac{(\lambda t)^1}{1!} + \frac{(\lambda t)^3}{3!} + \dots \right]$$

$$= e^{-\lambda t} \sinh \lambda t$$

$E[X(t)]$ is fn of t .

$\therefore X(t)$ is non-stationary.

(18). prove that the Random telegraph signal process $Y(t) = \alpha X(t)$ is a WSS process when α is RV which is independent of $X(t)$. Assume the values -1 and 1 which equal probability and $R_{XX}(t_1, t_2) = e^{-\lambda(t_1 - t_2)}$

Sol:-

$$X(t) = \alpha (-1)^{N(t)}$$

α	-1	1
$P(\alpha)$	$1/2$	$1/2$

$$E[X(t)] = E[\alpha (-1)^{N(t)}] = E[\alpha] E[(-1)^{N(t)}]$$
$$= E[\alpha] (-1)^{N(t)}$$

$$E[\alpha] = (-1) \times \frac{1}{2} + (1) \times \frac{1}{2} = 0.$$

$E[X(t)]$ is a constant.

$$R(t_1, t_2) = E[X(t_1) X(t_2)]$$
$$= E[\alpha^2 X(t_1) X(t_2)]$$
$$= E[\alpha^2] E[X(t_1) X(t_2)]$$
$$= 1 \cdot e^{-\lambda(t_1 - t_2)}$$
$$= e^{-\lambda(t_1 - t_2)}$$

$\therefore R$ is a fn of $t_1 - t_2$

$\therefore Y$ is a WSS process.

Proof:

sum of two independent poisson

process is a poisson process.

Sol:

$$P(X(t) = n) = \frac{e^{-\lambda t} (\lambda t)^n}{n!}; n = 0, 1, 2, \dots$$

with parameter λt :

$$X(t) = X_1(t) + X_2(t)$$